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Hearing in the ferret (*Mustela putorius*): thresholds for pure tone detection

Jack B. Kelly *, Gerard L. Kavanagh and James C.H. Dalton

Department of Psychology, Carleton University, Ottawa, Ont. K1S 5B6, Canada

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Thresholds for pure tone detection were determined for two male ferrets (*Mustela putorius*) using standard behavioral testing procedures. Both animals had a broad range of hearing comparable to that seen in other Carnivores. The lower limit of hearing at 60 dB SPL was estimated from the average audiogram to be 36 Hz, and the upper limit of hearing was 44 kHz. The lowest thresholds were found in the region of 8–12 kHz with both animals exhibiting sharp tuning at best frequency.

ferret, audiogram, hearing, absolute thresholds

Introduction

Over the past 10 years the ferret has been used extensively in a number of research areas including virology, physiology and immunology (see Frederick and Babish, 1985). Recently, the ferret has also been used in studies of the mammalian auditory system. In our laboratory we have shown that bilateral destruction of auditory cortex in the ferret results in severe impairments in sound localization. Deficits following unilateral lesions are confined exclusively to the auditory hemifield contralateral to the site of cortical damage (Kavanagh and Kelly, 1983, 1986). Similar results have been reported for other mammals including the cat, dog and monkey (Heffner, 1978; Heffner and Masterton, 1975; Jenkins and Masterton, 1982; Jenkins and Merzenich, 1984; Neff et al., 1956; Ravizza and Diamond, 1974; Strominger, 1969a,b; Thompson and Cortez, 1983; Wegener, 1964).

The response properties of neurons within the primary auditory cortex and the inferior colliculus of the ferret have also been examined using conventional microelectrode mapping techniques (Kelly et al., 1986; Moore et al., 1983). As in other

mammals, unit activity within these structures is characterized by short latency responses, sharp tuning and precise tonotopic organization. These studies indicate that the ferret is an excellent animal for further physiological investigation of the central auditory pathway.

As pointed out by Moore (1982), the ferret is also interesting from a developmental perspective because it is born in a very immature state following a gestation period of only 42 days. The eyes do not open until 30 days following birth. The late onset of visual function is reflected in the delayed postnatal development of the dorsal lateral geniculate (Linden et al., 1981) and the absence of visually evoked activity within cerebral cortex in immature ferrets (Rose, 1981). The onset of hearing in the ferret is also delayed. The ear canals do not open until the end of the first postnatal month. Startle responses to loud hand claps and acoustically evoked neural activity in the inferior colliculus occur soon thereafter (Moore, 1982). Thus, the ferret is an excellent candidate for early developmental manipulations which are difficult or impossible in other commonly used research animals.

The purpose of the present study was to provide baseline data for future studies with the ferret by determining the range and sensitivity of hearing in this species.

* To whom correspondence should be addressed.

Methods

Subjects

Behavioral thresholds for pure tone detection were determined using two pigmented adult male ferrets (*Mustela putorius*) obtained from Marshall Research Animals, North Rose, New York. Animals were housed individually in rack mounted cages (25 in. $\times$ 24 in. $\times$ 29 in.) made from 1 in.² wire mesh, and maintained on a diet of Purina cat chow. During training and testing animals were deprived of water in their home cage and given their entire daily requirement (approximately 45 ml) in the test apparatus. Otoscopic examinations were performed bi-weekly throughout the experiment, and in no instance was there any indication of obstruction, infection, or abnormality that could affect auditory sensitivity.

Behavioral apparatus

The behavioral apparatus was a semicircular enclosure with a radius of 3 feet. The floor was constructed of 1/2 in. plywood, and the sides and top were made of 1/2 in.² reinforced wire mesh. A 15 in. $\times$ 4 in. platform raised 1 in. from the floor was located at the center of the apparatus and served as a start area. The back third of the start area was covered with 1/8 in. copper plating, and a wire tunnel, 5 in. in diameter, was positioned over the center third to guide the animal's body position. An animal was able to initiate a trial by keeping its hind feet on the copper plating and touching its nose to a raised center spout located 2 in. in front of the start area. A 15 W light, positioned outside the apparatus, was used to signal the beginning of a trial. Auditory stimuli were presented from a loudspeaker located outside the test apparatus directly in front of the animal, i.e., at 0° azimuth. A 13 in. $\times$ 16 in. sound permeable barrier, consisting of vertically oriented stainless steel rods (1/8 in. in diameter) spaced 1/2 in. apart, was placed immediately in front of the center spout in order to prevent the animals from approaching the loudspeaker during stimulus presentation. Solenoid operated water spouts, located at -30 and +30° azimuth, were mounted on the wall of the apparatus 2 in. above the floor. Contact between a water spout and a stainless steel plate (10 in. wide) located on the floor around

the periphery of the apparatus provided the electrical circuit necessary for delivery of reinforcement.

The test apparatus was located in a sound-proof chamber (Eckel Industries) contained within a room separate from the experimenter and sound generating equipment. In order to reduce reflected sound, the walls and ceiling of the chamber were insulated with 2 in. fiberglass batting and draped with several layers of heavy burlap. A closed circuit television camera (Sony AVC 3200) mounted above the test apparatus was used to observe the animals' behavior during testing. Stimulus presentation, water delivery, and response tabulation were controlled by a microprocessor (Apple II) programmed in UCSD Pascal, and an interface designed by Carleton University's Science and Technology Center.

Sound generation

Pure tones were produced by passing a sine wave (Hewlett-Packard 3310A) through an electronic gate (Coulbourn S84-04), two bandpass filters (Krohn-Hite 3550), an attenuator (Hewlett-Packard 350D), and a power amplifier (Aiwa P30). Stimulus duration for each frequency was 540 ms including rise/fall times of 40 ms. Sounds were transduced by either a 30 cm woofer (Acoustic Research, Model 2 AX) for frequencies below 4 kHz, or a 10 cm piezoelectric tweeter (Tandy Electronics, Model 40-1382) for frequencies above 2 kHz. Speakers were located at ear level approximately 3.5 feet in front of the start area. In order to ensure that animals could not respond on the basis of vibratory cues, the loudspeaker used for low frequency testing was isolated from the behavioral apparatus and positioned on thick foam rubber pads. Under these conditions neither structurally transmitted nor airborne vibrations could be detected by human observers with ear plugs fitted securely in the external meatus. Sound pressure was determined by sampling a number of positions at the center spout using a measuring amplifier (Brüel and Kjaer 2608), a 1/2 in. condenser microphone (Brüel and Kjaer 4133), and a 1/3 octave bandpass filter set (Brüel and Kjaer 1615). The sound delivery system was capable of producing undistorted pure tones with no significant harmonics from 16 Hz to 64 kHz at an

intensity of at least 85 dB. The sound field within the start area was found to be relatively free of standing waves, and at no frequency did sound pressure levels at different positions vary by more than 6 dB. The intensity of ambient noise at the start area was also determined at 1/3 octave intervals throughout the measurable range, and found to be below the threshold for tone detection at each frequency.

Training

Animals were trained to enter the start area and initiate a trial by touching their nose to the center spout. Contact with the center spout triggered the presentation of either a 'silent' trial in which the trial-onset light alone was turned on, or a 'sound' trial in which a 540 ms noise burst (85 dB SPL) was presented in addition to the trial-onset light. The order of presentation, i.e. silent vs. sound trials, was determined according to a semi-random sequence. On silent trials (light alone) ferrets could receive a 0.3 ml water reinforcement by contacting the water spout at -30° azimuth, whereas on sound trials (light + auditory stimulus) they were required to contact the spout at $+30^\circ$ azimuth. Thus, animals were trained to go right ($+30^\circ$) when they heard a sound, and left (-30°) when they heard no sound. In the case of an incorrect response, the trial was terminated without reinforcement and the animals had to repeat the same trial and respond to the correct spout before reinforcement was given. These 'correction trials' were not included in the overall estimate of performance, as the stimulus condition, i.e., silence vs. sound, was not randomly determined. Training was considered complete when animals obtained in excess of 90% correct over the course of 50 randomized, i.e., noncorrection trials.

Psychophysical procedure

For psychophysical tests a 540 ms pure tone was substituted for the noise stimulus. Thresholds were obtained using a descending method of limits, in which the intensity of a tone was reduced in 10 dB steps until ferrets could no longer distinguish sound trials from silent trials. The general procedure was as follows. If an animal obtained 90% correct or better over the course of 50 trials, the intensity of the tone was reduced by 10 dB. If, at a

given attenuation, an animal obtained less than 90% correct, the intensity of the tone was not reduced and testing was continued for an additional 50 trials. Provided the two performance scores were in close agreement the intensity of the tone was reduced a further 10 dB. If, however, the scores were not in close agreement, sound pressure was held constant and testing was continued until performance stabilized. Thus, psychophysical curves for a given frequency were based on 50 trials for intensities at which animals obtained 90% correct or better, and at least 100 trials for intensities at which performance was below 90% correct. Performance scores were calculated from the percentage of correct responses following randomized trials, including both sound and no-sound conditions. A score of 50% indicated that an animal was unable to discriminate sound from silence. Thresholds were defined as the sound pressure level at which performance was statistically better than 50% according to the binomial distribution (i.e., 60% correct with a confidence limit of $P = 0.05$).

Results

Figure 1 shows the percentage of correct responses as a function of sound pressure for both ferrets at 62 Hz, 500 Hz, 4 kHz and 32 kHz. As can be seen, performance levels at high intensities were typically in excess of 90% correct, indicating that both animals were fully capable of making the appropriate behavioral response when tones were audible. As sound intensity was reduced both animals exhibited a steep decline in performance approximating 50% correct at the lowest sound pressure levels.

Individual audiograms for ferrets 97 and 98 are shown in Fig. 2. As can be seen, both animals had very good low frequency hearing, with a mean lower limit of 37 Hz at 60 dB SPL. At higher sound frequencies both animals showed a gradual increase in sensitivity up to 1 kHz. Thresholds remained low between 1 and 24 kHz, with narrowly tuned regions of maximum sensitivity in the 8–12 kHz range. Beyond 24 kHz sensitivity decreased gradually up to 32 kHz, and then more rapidly with an average interpolated upper limit of 44 kHz at 60 dB SPL.

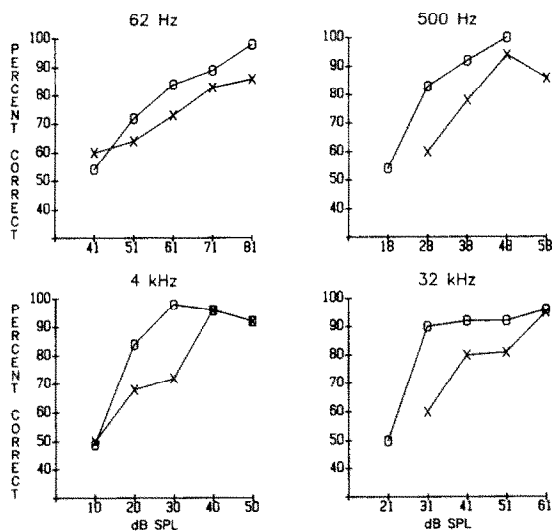


Fig. 1. Psychophysical curves for ferret 97 (O) and 98 (X) at representative sound frequencies. Performance levels are expressed as percent correct response and sound pressure levels are given in dB re 20 μ Pa.

Although thresholds for the two animals were generally in close agreement, some variability was noted in the middle frequency range, particularly at 8 kHz where thresholds were separated by 27 dB. This discrepancy remained despite the fact that animals were tested repeatedly at this frequency, with retest thresholds being within 4 dB of those obtained initially. In order to further explore the middle frequency range, additional thresholds were determined at 6, 10, 12 and 14

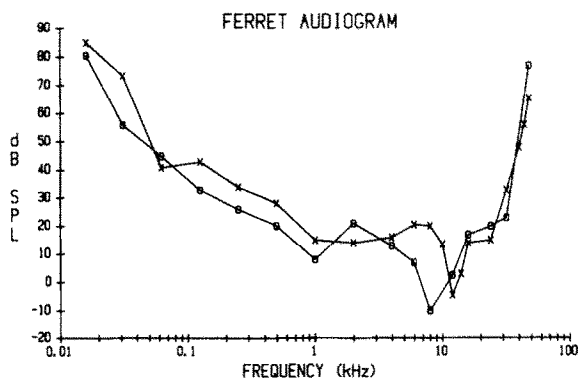


Fig. 2. Complete audiograms for ferret 97 (O) and 98 (X). Note the broad range of hearing with sharply turned areas of maximum sensitivity in the 8–12 kHz region.

kHz. These data points confirmed the observation that the two animals had reliably different best frequencies and exhibited narrow tuning of pure tone sensitivity in the 8–12 kHz region.

Discussion

Detailed comparisons can be made in Fig. 3 of the ferret's audiogram and audiograms from four other species in the order Carnivora, viz., dog, cat, raccoon and least weasel. A very close similarity can be seen between the audiograms of dogs and ferrets with regard to both the range of hearing and absolute sensitivity to sounds. Ferrets are somewhat more sensitive than dogs at the low frequency end of the audiogram, but for frequencies above 125 Hz the average absolute thresholds for these two species are within 10 dB of one another. The best sensitivity for both species is found within the 8–12 kHz range.

Comparison of the present results with those reported for the cat indicates that the ferret is more sensitive to frequencies below 125 Hz, but is generally less sensitive to higher frequencies. Our estimates of auditory sensitivity in the cat are based on data obtained from three separate studies by Heffner and Heffner (1985a), Neff and Hind (1955), and Elliott et al. (1960). These three studies present audiograms which differ as much as 25 dB in the mid-frequency range. The most recent investigation by Heffner and Heffner (1985a) shows that the sensitivity for mid-range

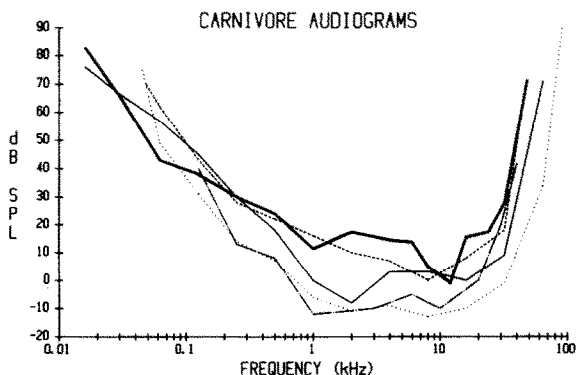


Fig. 3. Average ferret audiogram (heavy line) compared with four other Carnivores: Dog, *Canis familiaris* (dashed line); cat, *Felis catus* (dotted line); raccoon, *Procyon lotor* (dot-dash); and least weasel, *Mustela nivalis* (solid line).

frequencies is only slightly (about 15 dB) better than our estimates for the ferret, whereas the two older studies indicate significantly better hearing throughout this range. The high frequency limit of hearing for the ferret (44 kHz at 60 dB SPL) is clearly lower than the limit of 78 kHz reported for the cat (Heffner and Heffner, 1985a).

As can be seen in Fig. 3, a behavioral audiogram is also available for the raccoon. This animal, like the cat, is more sensitive than the ferret to mid-range frequencies. High frequency hearing, on the other hand, does not exceed that of the ferret.

A particularly interesting comparison can be made between the ferret and the least weasel. Both are members of the genus *Mustela* and are adapted to a similar lifestyle of predation on small rodents (Walker, 1968). However, they differ dramatically in body size with a typical male ferret weighing about 1000–1500 g, and a least weasel weighing between 50–70 g. Furthermore, the two species differ substantially in interaural distance, a factor which is closely related to hearing ability in mammals generally. As discussed originally by Masterton et al. (1969), and more recently by Heffner and Heffner (1982, 1983, 1985b), there is a very high negative correlation between interaural distance and the upper limit of hearing which is probably related to the availability of binaural cues for sound localization. Thus, one might expect the least weasel to have a higher upper limit of hearing than the ferret regardless of taxonomic proximity.

Generally, the auditory sensitivity of the least weasel and ferret is remarkably close. Both animals exhibit excellent low frequency sensitivity with a 60 dB low frequency cut-off of 36 Hz for the ferret, and 51 Hz for the least weasel. Indeed, they are both capable of detecting 16 Hz at about 80 dB SPL, a capacity which is not found in other Carnivores tested to date. Hearing in the mid-frequency range is excellent for both the ferret and least weasel, but the points of maximum sensitivity are not identical. The best frequency, 2 kHz, reported for the least weasel is well below the 8–12 kHz range found for the ferret. The least weasel also has a higher upper limit of hearing as might be expected from its smaller head size. The ferret's upper limit is 44 kHz compared to 60.5

kHz for the least weasel. It should be noted, however, that both values fall short of the expected value. A typical male ferret in our laboratory was found to have a separation between the external meatuses of 4.2 cm. With a sound source located at 90°, a distance of 5.4 cm would be required for the sound to travel around the head (Mills, 1972). Thus, in the ferret the estimated maximum binaural time difference is 156.6 μ s. From the correlational data published by Heffner and Heffner (1982, 1983), the expected upper limit associated with this time difference is 56 kHz rather than the 44 kHz actually obtained. From the same correlation, Heffner and Heffner (1985b) predicted an upper limit for the least weasel of 79.8 kHz rather than the observed limit of 60.5 kHz.

Electrophysiological data have been reported for both inferior colliculus, and primary auditory cortex of the ferret (Moore et al., 1983; Kelly et al., 1986). Estimates of auditory sensitivity based on responses of single neurons or neuron clusters correspond generally to the behavioral values reported here. For example, the most sensitive neurons in the central nucleus of the inferior colliculus or the primary auditory cortex are found in the frequency range between 8 and 12 kHz as expected (see Fig. 4). At these frequencies neurons respond to sound pressures which are near the absolute values obtained for behavioral thresholds. At lower frequencies there is a tendency for behavioral and electrophysiological data to diverge. This may be due in part to the limited sample size available for low frequency neurons. In our physiological mapping studies we have not encountered significant numbers of cells with best frequencies below about 0.2 kHz. The tuning curves of these neurons, however, extend to the low frequency limit of the ferret's behavioral audiogram. Unequivocal responses have been demonstrated to sound frequencies as low as 20 Hz.

An apparent discrepancy can be seen in Fig. 4 between behavioral and physiological data at extremely high frequencies. This difference, however, may be attributable to the fact that the high frequency representation in the ferret's primary auditory cortex is located at the extreme dorsal boundary of the ectosylvian gyrus, and may roll over into the suprasylvian sulcus to various de-

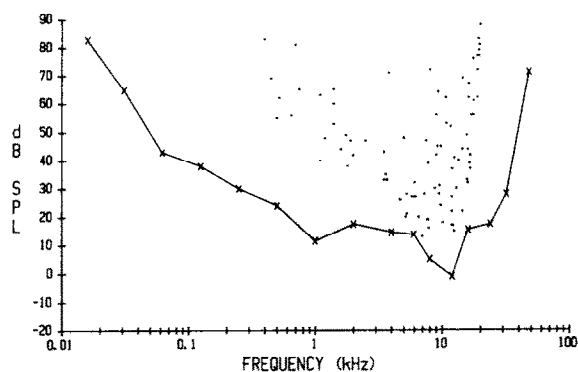


Fig. 4. Mean behavioral audiogram compared with thresholds at characteristic frequency for neurons in the ferret's primary auditory cortex (Kelly et al., 1986).

grees in individual animals. The depths of the suprasylvian sulcus have not yet been systematically investigated. A lack of high frequency responses reported by Moore et al. (1983) for neurons within the inferior colliculus was attributed by them to the inefficiency of their sound delivery system.

In summary, the ferret has been used to advantage in behavioral studies of the effects of cortical lesions on sound localization (Kavanagh and Kelly, 1983, 1986). Electrophysiological studies have shown the ferret to be well suited to microelectrode mapping studies of the central auditory pathway (Kelly et al., 1986; Moore et al., 1983). Also, because of its relative immaturity at birth, the ferret is likely to be a useful animal in studies of the development of the auditory system (Moore, 1982). The present study indicates that the ferret has excellent hearing over a broad range of frequencies, and is comparable in most respects to other carnivores which have been used as laboratory animals. These results encourage the continued use of the ferret in auditory research.

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